

Impact of Demographic Factors on Diet Diversity in U.S. Adults

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Introduction

A balanced diet is crucial for nutrition and overall health. Consuming a wide variety of food groups has been associated with improved gut health, lower health risks, and better long-term health. (Hossain et al., 2023) However, diet behaviours are shaped by numerous demographic factors. Using socioeconomic status as an example, individuals with higher income and education generally have higher food security and knowledge about maintaining a balanced diet. As a result, they have more access to a wide range of high-quality foods, such as fresh fruits, vegetables, and protein, resulting in a more diverse diet. In contrast, populations with lower socioeconomic status often face constraints such as food prices, access, and time, which limit dietary variety. influence diet quality and food consumption patterns.

A combination of other demographic factors, such as age and gender, also influences dietary behaviors. Dietary diversity remains difficult to predict because dietary behavior is highly variable, and many factors contribute to it. This research aims to study the association between demographic characteristics and dietary diversity among U.S adults and compare regression methods in predicting dietary diversity. To understand the relationship between demographic factors and diet diversity, dietary data were analyzed using linear regression. Least Absolute Shrinkage and Selection Operator (LASSO) regression and Random Forest were also performed to identify key factors that have the greatest impact on diet diversity.

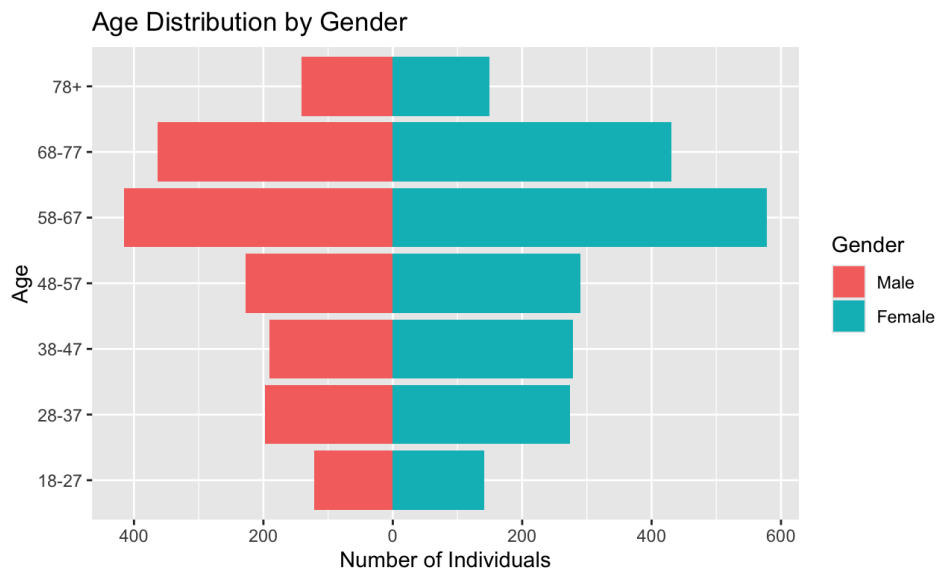
Methods and Materials

Data Summary

Data were obtained from the National Health and Nutrition Examination Survey (NHANES). Four datasets from the August 2021-August 2023 cycle– Dietary Interview - Individual Foods, First Day and Second Day (DR1IFF_L and DR2IFF_L), Demographics Data (DEMO_L), and Food Security Questionnaire (FSQ_L)– were combined based on participant ID (SEQN). Participants with complete dietary data for both days and non-missing values were included. The final dataset contained 3798 individuals of the following demographics. (See Fig.1a and 1b)

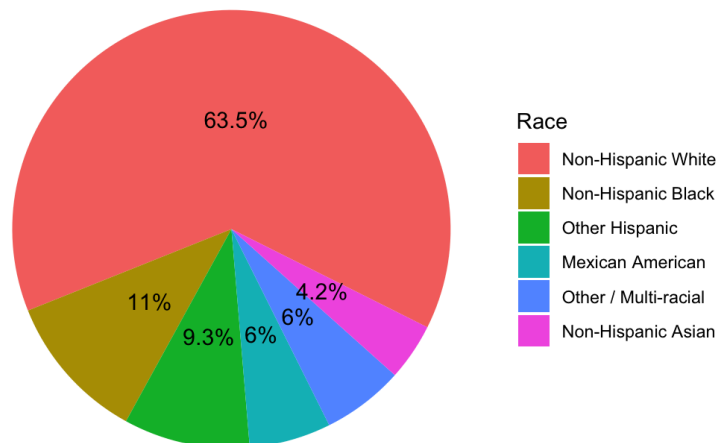
Figure 1. Age, Gender, and Race Distribution

a.



b.

Race Distribution



Explanatory Variables

- Age (RIDAGEYR_s)
- Gender (RIAGENDR)
- Level of education (DMDEDUC2)
- Income to poverty ratio (INDFMPIR_s)
- Race (RIDRETH3)
- Country of Birth (DMDDBORN4)
- Household size (DMDHHSIZ_s)
- Marital status (DMDMARTZ)
- Level of food security (FSDAD)

Categorical variables were converted into factors for analysis. Continuous variables, including age, household size, and income-to-poverty ratio, were standardized because the LASSO regression is not scale invariant and were used in all methods for comparability.

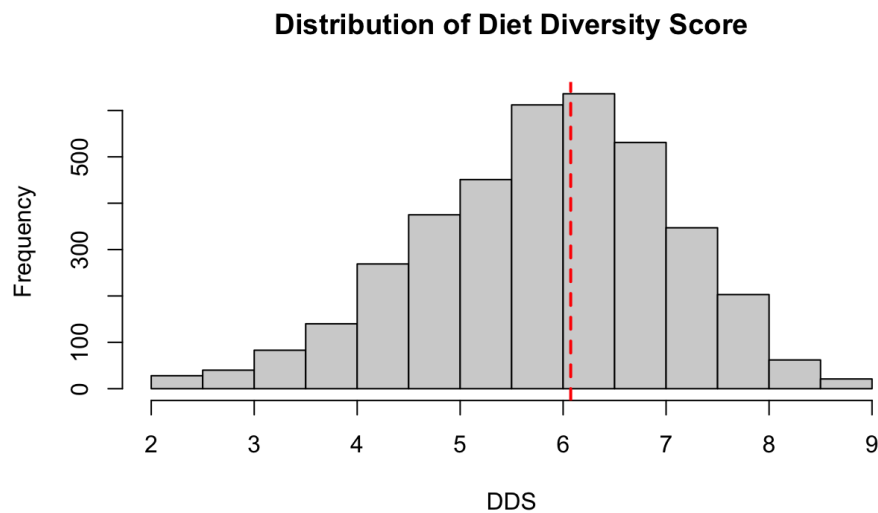
Response Variable

Dietary diversity was measured using a diet diversity score (DDS), which counts the number of food groups consumed within a day. DDS is commonly used as a proxy for measuring nutritional adequacy. (Zhao et al., 2020) A higher DDS indicates greater dietary diversity. In this analysis, DDS is calculated using the USDA food code variable (DR1IFDCD) in both days of dietary interview. The first digit of the food code categorizes the food item into one of nine major food groups. The number of unique food groups was summarized for each day, then averaged for each participant as the final response.

$$DDS = (DDS_{DAY1} + DDS_{DAY2}) / 2$$

DDS ranged from 2 to 9 food groups, with an approximately normal distribution centered around a mean DDS of 6. (See Fig.2)

Figure 2. DDS distribution with mean



Statistical Methods

Summary statistics were used to analyze the relationships between DDS and the demographic variables. Histograms and diagnostic plots were used to assess assumptions.

The data was randomly divided into training and validation sets, using an 80-20 split to evaluate the prediction accuracy across models. The full prediction model was fitted on the training set.

Full Prediction Model:

$$DDS \sim \text{Level of education} + \text{Gender} + \text{Age} + \text{Country of birth} + \text{Marital status} + \text{Household size} + \text{Income to poverty ratio} + \text{Race} + \text{Level of food security}.$$

Since DDS was calculated as the average number of food groups consumed across two days and followed an approximately continuous distribution, a multiple linear regression model was fitted instead of Poisson regression, typically used for count data. Histograms and diagnostic plots were used to assess assumptions. Variance Inflation Factors (VIF) were also computed with `cars::vif()` to determine any multicollinearity among predictors. LASSO regression, which improves model accuracy by imposing a shrinking penalty, was also performed to eliminate less important demographic variables as the predictor coefficients are set to zero. Using cross-validation, the optimal penalty (λ) was selected. Additionally, Random Forest, which uses multiple regression trees to improve accuracy, was fitted. It was conducted to assess potential nonlinear relationships and to understand variable importance using advanced machine learning approaches. It is included to compare the performance between more flexible models and simple linear regression.

Training R^2 values and test root mean square error (RMSE) were used to compare model performance between the three statistical methods.

Results

Linear Regression

The multiple linear regression fitted on the full prediction model explained 16.82% of the variation in DDS. ($R^2 = 0.1682$) with a test RMSE of 1.124. The low level of explanation is expected as dietary behaviours are highly variable and driven by complex factors. Diagnostic plots above (See Fig. 5) indicate that the model has met all assumptions. DDS follows an approximately normal, linear distribution with equal variance. Patterns observed in scale location could be explained by DDS being averages of discrete counts, hence having 0.5 increments. There are no significant outliers. GVIF values were below 2; no multicollinearity was detected (See Fig.4).

Most demographic variables show a significant association with diet diversity. Education, age, country of birth, income, race, and food security are all significant predictors of DDS. As adults obtain higher levels of education, their diet diversity increases, especially for college graduates. Older adults also have greater diet diversity. Non-US-born adults consume more diverse diets than those born in the US. Females generally have more diverse diets than males. Meanwhile, adults with low levels of food security and low income are significantly associated with a less diverse diet. Participants who are not married have less diverse diets. For differences between race, Non-Hispanic Asian and Black groups have significantly less diverse diets compared to Mexicans. Household size is not a significant predictor. (See Fig.3)

Figure 3. Linear Regression Output

Residuals:

Min	1Q	Median	3Q	Max
-3.8115	-0.7375	0.0292	0.7971	3.3352

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.77488	0.14966	38.587	< 2e-16	***
DMDEDUC29-11th grade	0.20351	0.14102	1.443	0.149090	
DMDEDUC2High school graduate/GED	0.31178	0.12938	2.410	0.016017	*
DMDEDUC2Some college or AA	0.41875	0.12751	3.284	0.001035	**
DMDEDUC2College graduate+	0.68977	0.13092	5.269	1.47e-07	***
RIAGENDRFemale	0.10916	0.04247	2.570	0.010208	*
DMDBORN4Non-US	0.26102	0.07062	3.696	0.000223	***
DMDMARTZWidowed/Divorced/Separated	-0.17722	0.05481	-3.233	0.001236	**
DMDMARTZNever married	-0.12496	0.06443	-1.940	0.052523	.
RIDRETH3Other Hispanic	-0.10579	0.11232	-0.942	0.346361	
RIDRETH3Non-Hispanic White	-0.16242	0.10212	-1.590	0.111843	
RIDRETH3Non-Hispanic Black	-0.26354	0.11352	-2.322	0.020322	*
RIDRETH3Non-Hispanic Asian	-0.33626	0.13727	-2.450	0.014354	*
RIDRETH3Other / Multi-racial	-0.17033	0.12604	-1.351	0.176670	
FSDADMarginal	-0.04541	0.07599	-0.598	0.550149	
FSDADLow	-0.21718	0.07934	-2.737	0.006232	**
FSDADVery low	-0.35531	0.08452	-4.204	2.70e-05	***
RIDAGEYR_s	0.28492	0.02475	11.510	< 2e-16	***
INDFMPIR_s	0.18408	0.02722	6.762	1.62e-11	***

DMDHHSIZ_s 0.01283 0.02457 0.522 0.601534

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.135 on 3018 degrees of freedom

Multiple R-squared: 0.1682, Adjusted R-squared: 0.163

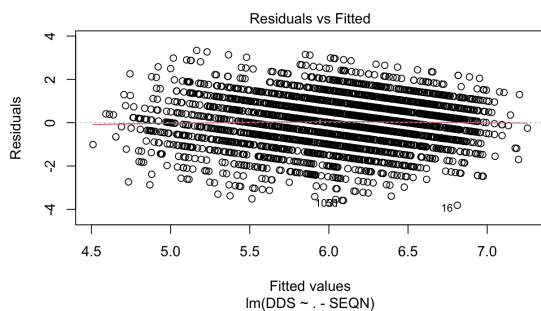
F-statistic: 32.12 on 19 and 3018 DF, p-value: < 2.2e-16

Figure 4. Variance Inflation Factor

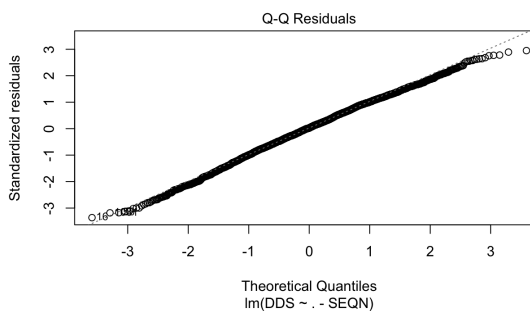
	GVIF	Df	GVIF^(1/(2*Df))
DMDDEDUC2	1.713035	4	1.069599
RIAGENDR	1.045464	1	1.022479
DMDBORN4	1.714270	1	1.309301
DMDMARTZ	1.680449	2	1.138561
RIDRETH3	2.072440	5	1.075594
FSDAD	1.525376	3	1.072909
RIDAGEYR_s	1.448465	1	1.203522
INDFMPIR_s	1.739582	1	1.318932
DMDHHSIZ_s	1.412973	1	1.188685

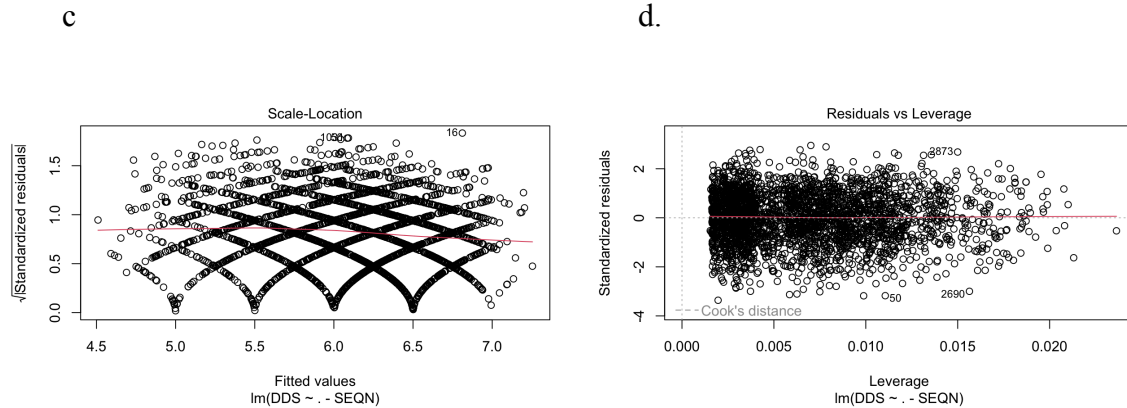
Figure 5. Diagnostic Plots

a.



b.





LASSO Regression

Cross-validation selected an optimal penalty of $\lambda = 0.009293$. LASSO Regression explained 16.52% of the variation. It had a test RMSE of 1.131487, slightly higher than linear regression.

LASSO Regression did not improve predictive accuracy compared to linear regression, but helped remove weaker estimates. Some levels of categories were shrunk to zero, including adults who completed education through 9-11th grade, Hispanics other than Mexicans, Non-Hispanic whites, and other/multiracial adults. Education, age, country of birth, income, and food security remained among the strongest predictors of dietary diversity.

Figure 6. LASSO Regression Output

	Estimate
(Intercept)	5.927791487
DMDEDUC29-11th grade	-0.049120854
DMDEDUC2High school graduate/GED	.
DMDEDUC2Some college or AA	0.098939918
DMDEDUC2College graduate+	0.371004853
RIAGENDRFemale	0.088179985
DMDBORN4Non-US	0.230230723
DMDMARTZWidowed/Divorced/Separated	-0.155182413
DMDMARTZNever married	-0.119524898
RIDRETH3Other Hispanic	.
RIDRETH3Non-Hispanic White	.
RIDRETH3Non-Hispanic Black	-0.092495741
RIDRETH3Non-Hispanic Asian	-0.108979327

RIDRETH3Other / Multi-racial	.
FSDADMarginal	-0.016206781
FSDADLow	-0.187903950
FSDADVery low	-0.332141680
RIDAGEYR_s	0.263691288
INDFMPIR_s	0.189595457
DMDHHSIZ_s	0.001737936

Random Forest

Call:

```
randomForest(formula = DDS ~ . - SEQN - RIDAGEYR - INDFMPIR - DMDHHSIZ,
data = train, mtry = 3, importance = T)
```

Type of random forest: regression

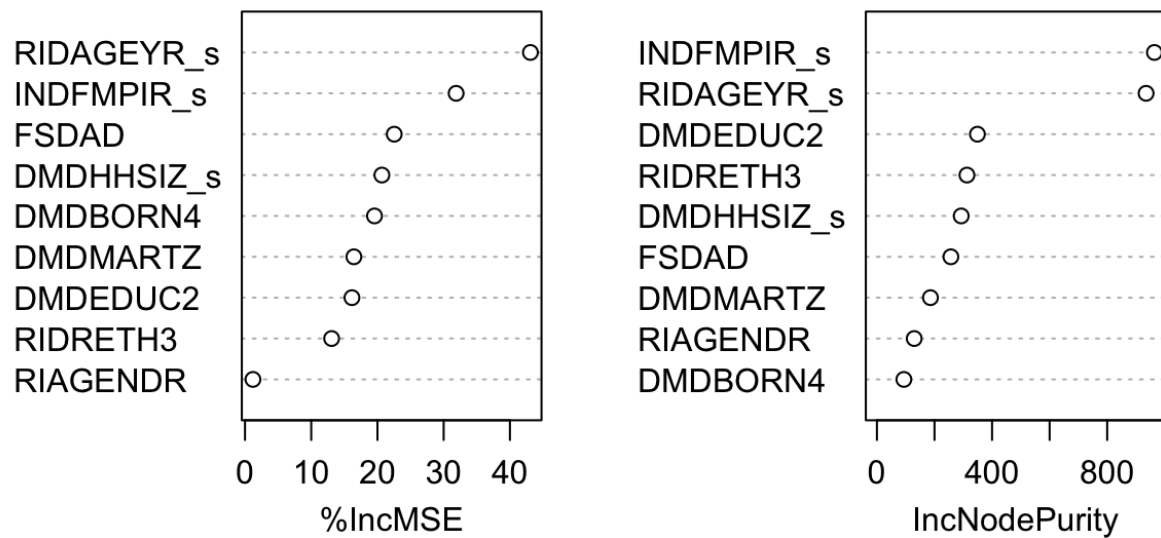
Number of trees: 500

No. of variables tried at each split: 3

Mean of squared residuals: 1.361036

% Var explained: 11.51

Figure 7. Random Forest Variable Importance Plot



For Random Forest, only 11.51% of the variance in DDS is explained, and the test RMSE is 1.155613. Random Forest performed worse than both linear regression and LASSO regression.

Income and age are shown to be the most important variables in predicting diet diversity. This is consistent with both regression approaches, which showed both as the most significant variables in predicting DDS. However, gender showed low importance in predicting DDS.

Although advanced and more flexible methods, such as LASSO regression and Random Forest, are often expected to improve prediction accuracy and explain a higher proportion of variance than linear regression, multiple linear regression had the lowest RMSE overall. Due to the linear nature of the DDS score in the data and the variability of diet diversity, Random Forest performed the worst. Low multicollinearity among predictors also reduced the benefits of regularization methods like LASSO.

Discussion

This study demonstrates that socioeconomic and demographic factors are strongly associated with dietary diversity in U.S. adults. Higher education, income, and food security are associated with greater dietary diversity across all methods. This reflects the importance of socioeconomic status in increasing access to a variety of nutritious foods. Conversely, lower socioeconomic status is associated with reduced dietary diversity, likely due to financial and environmental constraints.

Overall, age is most influential in predicting diet diversity besides socioeconomic status. Older adults generally have more diverse diets. Non-US-born participants also showed greater dietary diversity, which may reflect cultural dietary differences and consumption of a wider variety of traditional foods. Although race and gender were associated with DDS, their effects were smaller compared to age, income, and education. Public policies and programs should aim to improve diet diversity by promoting affordable and accessible foods to reduce socioeconomic disparities for groups with lower DDS.

All three models had RMSE values close to 1, indicating that DDS predictions were within about one food group of the observed diet diversity. The comparison between different statistical methods also suggested reduced advantages of more flexible models, especially because of the linearity of this dietary data.

Several limitations should be considered, as data may not be representative of the population. Dietary interviews were self-reported and are prone to recall bias and human errors. Since only two days of data were collected, it does not portray long-term dietary behavior and diversity. Further work includes using more complex DDS to better quantify diet diversity and examine nutritional quality. Different approaches in predicting diet diversity should also be explored, such as using interaction terms to improve model accuracy.

References

- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: With applications in R. Springer.
- Centers for Disease Control and Prevention. Dietary analyses tutorials. *National Health and Nutrition Examination Survey (NHANES)*.
<https://wwwn.cdc.gov/nchs/nhanes/tutorials/DietaryAnalyses.aspx>
- Centers for Disease Control and Prevention. NHANES 2021–2023 datasets and documentation. National Health and Nutrition Examination Survey (NHANES).
<https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023>
- Zhao, W., Zhang, J., Zhao, A., Wang, P., & Zhang, Y. (2020). Dietary diversity scores: An indicator of micronutrient inadequacy instead of obesity for Chinese children. *BMC Public Health*, 20, 462. <https://doi.org/10.1186/s12889-020-8451-7>
- de Oliveira Santos, M., et al. (2024). Dietary diversity and associated socioeconomic factors among adults: A population-based analysis. *Nutrients*. Advance online publication.
<https://doi.org/10.3390/nu16193334>
- Hossain, M. B., et al. (2023). Dietary diversity and nutritional adequacy among adults: Evidence from population studies. *Nutrients*, 15(9), 2154. <https://doi.org/10.3390/nu15092154>

Appendix

Additional Figures

